Clock Cycle (Presses)	000 Instr (IF Stage)	100 Ext_Imm (ID Stage)	101 ALURes (EX Stage)	111 Write Data (WB Stage)	
0 (Start)	2085 (ADDI)	0	----	----	ADDI $1 enters the Fetch stage.
1	0000 (NoOp)	5	0	----	ADDI $1 --> Decode. It pulls the 5 from the instruction.
2	0000 (NoOp)	0	5	0	ADDI $1 --> Execute. ALU adds 0 + 5.
3	0000 (NoOp)	0	0	0	ADDI $1 --> Memory. (Nothing happens for ADDI).
4	2502 (ADDI $2)	0	0	5	ADDI $1 --> Write Back, It safely writes 5 to $1. Meanwhile, next inst ADDI $2 enters Fetch.
5	0000 (NoOp)	2	0	0	ADDI $2 --> Decode. Because it waited, it reads $1 = 5 and Ext_Imm = 2.
6	0000 (NoOp)	0	7	0	ADDI $2 --> Execute. ALU calculates 5 + 2 = 7.
7	0000 (NoOp)	0	0	0	ADDI $2 --> Memory.
8	0530 (ADD $3)	0	0	7	ADDI $2 --> Write Back and writes 7 to $2.
9	0000 (NoOp)	0	0	0	ADD $3 --> Decode. Reads $1=5 and $2=7 from the reg (switch: 010 and 011 to see 5 and 7)
10	0000 (NoOp)	0	000C	0	ADD $3 --> Execute. ALU calculates 5 + 7 = 12 (000C in Hex).
11	0000 (NoOp)	0	0	0	ADD $3 --> Memory.
12	E00C (J 12)	0	0	000C	ADD $3 --> Write Back and saves C. Jump inst enters Fetch.
13	0000 (NoOp)	0	0	0	J 12 is in Decode. The main control unit reads the jump and forces the PC back to 12.
14	E00C (J 12)	0	0	0	Because of the jump, the processor fetches J 12 again --> infinite loop

Instr \ Clk	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
ADDI $1, $0, 5	IF	ID	EX	MEM	WB												
NoOp		IF	ID	EX	MEM	WB											
NoOp			IF	ID	EX	MEM	WB										
NoOp				IF	ID	EX	MEM	WB									
ADDI $2, $1, 2					IF	ID	EX	MEM	WB								
NoOp						IF	ID	EX	MEM	WB							
NoOp							IF	ID	EX	MEM	WB						
NoOp								IF	ID	EX	MEM	WB					
ADD $3, $1, $2									IF	ID	EX	MEM	WB				
NoOp										IF	ID	EX	MEM	WB			
NoOp											IF	ID	EX	MEM	WB		
NoOp												IF	ID	EX	MEM	WB	
J 12													IF	ID	EX	MEM	WB